

Question:

I have 10 liters of orange drink that are two-thirds water and I wish to add it to 15 liters of pineapple drink that is three-fifths water. But as I pour it, I spill one liter of the orange drink. How much water is in the remaining 24 liters?

Reasoning + Answer:

Let's break down the problem step by step to determine how much water is in the remaining 24 liters after adding the drinks and accounting for the spill .

1 . **Calculate the amount of water in the original 10 liters of orange drink : **

- The orange drink is two -thirds water .
- Let (W_{orange}) represent the amount of water in the orange drink .
- We can set up the equation :

$$W_{\text{orange}} = \frac{2}{3} \times 10$$

- Perform the multiplication :

$$W_{\text{orange}} = \frac{2 \times 10}{3} = \frac{20}{3} \approx 6.67 \text{ liters }$$

2 . **Calculate the amount of water spilled : **

- One liter of the orange drink is spilled .
- We need to determine what fraction of this liter is water .
- Since the original orange drink was two -thirds water , the spilled liter also contains two -thirds water .
- Let (W_{spilled}) represent the amount of water in the spilled liter .
- We can set up the equation :

$$W_{\text{spilled}} = \frac{2}{3} \times 1$$

- Perform the multiplication :

$$W_{\text{spilled}} = \frac{2}{3} \approx 0.67 \text{ liters }$$

3 . **Calculate the remaining amount of water in the orange drink after the spill : **

- Subtract the amount of water spilled from the original amount of water in the orange drink .
- Let $(W_{\text{remaining_orange}})$ represent the remaining amount of water in the orange drink .
- We can set up the equation :

$$W_{\text{remaining_orange}} = W_{\text{orange}} - W_{\text{spilled}}$$

- Substitute the known values :

$$W_{\text{remaining_orange}} = \frac{20}{3} - \frac{2}{3}$$

- Perform the subtraction :

$$W_{\text{remaining_orange}} = \frac{20 - 2}{3} = \frac{18}{3} = 6 \text{ liters }$$

4 . **Calculate the amount of water in the 15 liters of pineapple drink : **

- The pineapple drink is three -fifths water .
- Let $(W_{\text{pine apple}})$ represent the amount of water in the pineapple drink .
- We can set up the equation :

$$W_{\text{pine apple}} = \frac{3}{5} \times 15$$

- Perform the multiplication :

$$W_{\text{pine apple}} = \frac{3 \times 15}{5} = \frac{45}{5} = 9 \text{ liters }$$

5 . **Calculate the total amount of water in the remaining 24 liters : **

- Add the remaining amount of water in the orange drink to the amount of water in the pineapple drink .
- Let (W_{total}) represent the total amount of water in the mixture .
- We can set up the equation :

$$W_{\text{total}} = W_{\text{remaining_orange}} + W_{\text{pine apple}}$$

- Substitute the known values :

$$W_{\text{total}} = 6 + 9 = 15 \text{ liters }$$

Thus , the amount of water in the